

Phase–Amplitude Equations

Organized transcription of four handwritten pages

Editorial convention. A dot denotes $d/d\tau$. Derivative primes in the original are replaced by dots; parameter primes, such as κ' and γ' , are retained. Crossed-out intermediate text is omitted. Some symbol readings are provisional; see the transcription notes on page 3.

1 Angular and radial equations

For the first page, write $\mathbf{r} = u\hat{\mathbf{x}} + v\hat{\mathbf{y}}$, with $u = r \cos \theta$, $v = r \sin \theta$, and $r^2 = u^2 + v^2$. The angular derivative is

$$\dot{\theta} = \frac{\frac{d}{d\tau}(v/u)}{1 + (v/u)^2} = \frac{u^2 \frac{\dot{v}u - \dot{u}v}{u^2}}{u^2 + v^2} = \frac{\dot{v}u - \dot{u}v}{u^2 + v^2}. \quad (1)$$

The transverse vector equation in the notes reads

$$\begin{aligned} \frac{d}{d\tau}(u\hat{\mathbf{x}} + v\hat{\mathbf{y}}) &= -[(u\hat{\mathbf{x}} + v\hat{\mathbf{y}}) \times \hat{\mathbf{z}} + \kappa'(\chi - 1)\hat{\mathbf{z}} \times \mathbf{b}'] \\ &= -[-u\hat{\mathbf{y}} + v\hat{\mathbf{x}} + \kappa'\kappa'(\chi - 1)(\cos \theta_B \hat{\mathbf{y}} - \sin \theta_B \hat{\mathbf{x}})]. \end{aligned} \quad (2)$$

Thus

$$\begin{cases} \dot{u} = -v + \kappa'(\chi - 1) \sin \theta_B, \\ \dot{v} = u - \kappa'(\chi - 1) \cos \theta_B. \end{cases} \quad (3)$$

With $\delta = \theta_B - \theta$,

$$\begin{aligned} \dot{v}u - \dot{u}v &= v^2 + u^2 - \kappa'(\chi - 1)r(\sin \theta_B \sin \theta + \cos \theta_B \cos \theta) \\ &= r^2 - \kappa'(\chi - 1)r \cos \delta, \end{aligned} \quad (4)$$

so that

$$\dot{\theta} = 1 - \frac{\kappa'(\chi - 1) \cos \delta}{r}. \quad (5)$$

The handwritten reference “from Eq. (23)” gives

$$\begin{aligned} \dot{r} &= \hat{\mathbf{e}}_r \cdot \frac{d\mathbf{r}}{d\tau} = -\hat{\mathbf{e}}_r \cdot [\mathbf{r} \times \hat{\mathbf{z}} + \kappa'(\chi - 1)\hat{\mathbf{z}} \times \mathbf{b}'] \\ &= -\kappa'(\chi - 1)(\hat{\mathbf{z}} \times \mathbf{b}') \cdot \hat{\mathbf{e}}_r \\ &= -\kappa'(\chi - 1)(\mathbf{b}' \times \hat{\mathbf{e}}_r) \cdot \hat{\mathbf{z}} \\ &= \kappa'(\chi - 1) \sin \delta. \end{aligned} \quad (6)$$

2 Relative phase and coupled equations

The wave-phase derivative is transcribed as follows. The middle factors in this particular line are less clear in the photograph.

$$\begin{aligned}
 \dot{\theta}_B &= \frac{d}{d\tau}(\kappa' z') = -\kappa' \beta'_z \frac{\gamma'}{\Omega} c \\
 &= -\frac{k}{\gamma_T} \beta'_z \frac{\gamma'}{\Omega} c = -\frac{\omega}{c} \frac{m \gamma'}{\Omega \gamma_T} c \beta'_z \\
 &= -\alpha \gamma' \beta'_z = -(\chi - 1).
 \end{aligned} \tag{7}$$

Consequently,

$$\begin{aligned}
 \dot{\delta} &= \dot{\theta}_B - \dot{\theta} \\
 &= -(\chi - 1) - 1 + \frac{\kappa'(\chi - 1) \cos \delta}{r} \\
 &= -\chi + \frac{\kappa'(\chi - 1) \cos \delta}{r}.
 \end{aligned} \tag{8}$$

The three equations collected at the top of the second page are

$$\begin{aligned}
 \dot{\delta} &= -\chi + \frac{\kappa'(\chi - 1) \cos \delta}{r}, \\
 \dot{r} &= \kappa'(\chi - 1) \sin \delta, \\
 \dot{\chi} &= -\alpha \gamma' \varepsilon - r \kappa' \sin \delta.
 \end{aligned} \tag{9}$$

3 Eliminating the explicit phase term

The stated objective is to cancel the phase term on the right-hand side. First,

$$\begin{aligned}
 \dot{\chi} - r \dot{r} &= -\alpha \gamma' \varepsilon - r \kappa' \sin \delta - r \kappa' \chi \sin \delta + r \kappa' \sin \delta \\
 &= -\alpha \gamma' \varepsilon - r \kappa' \chi \sin \delta.
 \end{aligned} \tag{10}$$

The notes propose adding $d(\kappa' r \cos \delta)/d\tau$. Indeed,

$$\begin{aligned}
 \frac{d}{d\tau}(r \cos \delta) &= \dot{r} \cos \delta - r \sin \delta \dot{\delta} \\
 &= \kappa'(\chi - 1) \sin \delta \cos \delta - r \sin \delta \left(-\chi + \frac{\kappa'(\chi - 1) \cos \delta}{r} \right) \\
 &= \kappa'(\chi - 1) \sin \delta \cos \delta + r \chi \sin \delta - \kappa'(\chi - 1) \sin \delta \cos \delta \\
 &= \chi r \sin \delta.
 \end{aligned} \tag{11}$$

For constant κ' , this yields

$$\dot{\chi} - r \dot{r} + \frac{d}{d\tau}(\kappa' r \cos \delta) = -\alpha \gamma' \varepsilon, \tag{12}$$

or equivalently

$$\frac{d}{d\tau} \left(\chi - \frac{r^2}{2} + \kappa' r \cos \delta \right) = -\alpha \gamma' \varepsilon. \tag{13}$$

4 Phase-based Cartesian variables and J

The second page now introduces

$$u = r \cos \delta, \quad v = r \sin \delta. \quad (14)$$

These reuse the letters u, v for components defined by δ , whereas Section 1 used components defined by θ . Substitution gives

$$\frac{d}{d\tau} \left(\chi - \frac{u^2 + v^2}{2} + \kappa' u \right) = -\alpha \gamma' \varepsilon. \quad (15)$$

Introduce

$$J = \chi - \frac{u^2 + v^2}{2} + \kappa' u = \chi - \frac{(u - \kappa')^2 + v^2}{2} + \frac{\kappa'^2}{2}. \quad (16)$$

The handwritten integration is

$$\frac{dJ}{d\tau} = -\alpha \gamma' \varepsilon \implies J = J_0 - \alpha \gamma' \varepsilon \tau. \quad (17)$$

This displayed linear form presumes constant $\gamma' \varepsilon$ and $J_0 = J(0)$.

Finally, differentiating the variables in Eq. (14) gives

$$\frac{du}{d\tau} = \dot{r} \cos \delta - r \sin \delta \dot{\delta} = \chi r \sin \delta = \chi v, \quad (18)$$

$$\begin{aligned} \frac{dv}{d\tau} &= \dot{r} \sin \delta + r \cos \delta \dot{\delta} \\ &= \kappa'(\chi - 1) - u\chi = -\chi(u - \kappa') - \kappa'. \end{aligned} \quad (19)$$

Hence the final pair on the second page is

$$\dot{u} = \chi v, \quad \dot{v} = -\chi(u - \kappa') - \kappa'. \quad (20)$$

Transcription notes for the first two pages

1. The handwritten variable resembling x is typeset as χ . The primed factor in $-\alpha \gamma' \varepsilon$ is read as γ' ; its handwriting also resembles ν' . These are provisional symbol readings.
2. In the expanded $\dot{\theta}_B$ line, Ω_λ , m , and γ_T are provisional readings of the handwriting. Their definitions are not supplied in these photographs. This line is transcribed, not independently validated.
3. Overwritten signs in $\dot{v}u - \dot{u}v$ and $\dot{\theta}$ are rendered as minus signs, consistent with the immediately preceding component equations and the subsequent $\dot{\delta}$ equation.
4. The original reference to Eq. (23) is preserved as a reference in the handwritten source; that equation itself was not supplied separately.
5. The two definitions of u, v are intentionally retained and identified. The assumption that κ' is constant, and the condition for linear $J(\tau)$, are editorial clarifications. No definitions of the physical parameters have been added.
6. Partially visible writing on adjacent pages, crossed-out text, and unrelated marginal fragments are excluded.

5 Shifted variables and the complex equation

The third handwritten page continues with the shifted variables

$$q = u - \kappa', \quad p = v. \quad (21)$$

For constant κ' , the final equations of Section 4 become

$$\boxed{\dot{q} = \chi p, \quad \dot{p} = -\chi q - \kappa'.} \quad (22)$$

Set $z = q + ip$ and combine the two real equations:

$$\begin{aligned} \dot{q} + i\dot{p} &= \chi p - i\chi q - i\kappa', \\ \dot{z} &= -i\chi(q + ip) - i\kappa' = -i\chi z - i\kappa'. \end{aligned} \quad (23)$$

6 Introducing the detuning parameter Λ

Using the previous expression for J ,

$$J = \chi - \frac{q^2 + p^2}{2} + \frac{\kappa'^2}{2}, \quad \Lambda = \frac{q^2 + p^2}{2} - \chi. \quad (24)$$

It follows that

$$\begin{aligned} \Lambda &= \frac{\kappa'^2}{2} - J = \frac{\kappa'^2}{2} - (J_0 - \alpha\gamma'\varepsilon\tau) \\ &= \frac{\kappa'^2}{2} - J_0 + \alpha\gamma'\varepsilon\tau = \Lambda_0 + \alpha\gamma'\varepsilon\tau, \quad \Lambda_0 = \frac{\kappa'^2}{2} - J_0. \end{aligned} \quad (25)$$

Therefore $\chi = -\Lambda + (q^2 + p^2)/2 = -\Lambda + |z|^2/2$. Substitution into Eq. (23) yields

$$\dot{z} = -i \left(\frac{|z|^2}{2} - \Lambda \right) z - i\kappa' \implies \boxed{i\dot{z} - \left(\frac{|z|^2}{2} - \Lambda \right) z = \kappa'.} \quad (26)$$

7 Comparison with a nonlinear Schrödinger-type equation

The reference form written in the notes is

$$i \frac{d\Psi}{d\tau} + (|\Psi|^2 - \tau)\Psi = \mu, \quad \mu \in \mathbb{R}. \quad (27)$$

Taking the complex conjugate gives

$$-i \frac{d\Psi^*}{d\tau} + (|\Psi|^2 - \tau)\Psi^* = \mu. \quad (28)$$

For $z = -\Psi^*$, the last handwritten line reads

$$\frac{dz}{d\tau} + (\tau - |\Psi|^2)z = \mu \quad (\text{as written in the photograph}). \quad (29)$$

Editorial correction: direct substitution retains a factor i in front of the derivative. Thus the corrected line is

$$\boxed{i \frac{dz}{d\tau} + (\tau - |\Psi|^2)z = \mu, \quad |z|^2 = |\Psi|^2.} \quad (30)$$

Scope of the comparison: the two complex equations still have different coefficients ($|z|^2/2$ versus $|z|^2$, and $\Lambda(\tau)$ versus τ). The additional handwritten page supplies the scaling in Section 8 below. As before, linear $\Lambda(\tau)$ assumes constant κ' and $\gamma'\varepsilon$.

8 Normalization and the threshold written in the notes

The fourth handwritten page starts from

$$i \frac{dz}{d\tau} - \left(\frac{|z|^2}{2} - \Lambda \right) z = \kappa', \quad \Lambda = \Lambda_0 + \alpha \gamma' \varepsilon \tau. \quad (31)$$

The proposed substitutions are $z = AZ$, $\tau = BT$, and $\Lambda = CT$. With constant, real A and B , these give

$$i \frac{dZ}{dT} - Z \left(\frac{A^2 B}{2} |Z|^2 - BCT \right) = \frac{\kappa' B}{A}. \quad (32)$$

Choose

$$\frac{A^2 B}{2} = 1, \quad BC = 1, \quad a = \alpha \gamma' \varepsilon > 0, \quad C = \sqrt{a}. \quad (33)$$

Then the scaling factors in the notes are

$$B = \frac{1}{\sqrt{a}}, \quad A = (2\sqrt{a})^{1/2} = \sqrt{2} a^{1/4}, \quad \frac{A}{B} = \sqrt{2} a^{3/4}. \quad (34)$$

The dimensionless forcing and the resulting equation are

$$\mu = \frac{\kappa'}{A/B} = \frac{\kappa'}{\sqrt{2} a^{3/4}}, \quad \boxed{i \frac{dZ}{dT} - (|Z|^2 - T)Z = \mu.} \quad (35)$$

Time-origin clarification. The simultaneous choices $\tau = BT$ and $\Lambda = CT$ require $\Lambda_0 = 0$. For nonzero Λ_0 , retain the constant through a shift of the time origin:

$$\boxed{T = \sqrt{a} \left(\tau + \frac{\Lambda_0}{a} \right), \quad \tau = \frac{T}{\sqrt{a}} - \frac{\Lambda_0}{a}, \quad \Lambda = \sqrt{a} T.} \quad (36)$$

This shift leaves $dT/d\tau = \sqrt{a}$ unchanged, so the amplitude scaling and the normalized equation above still apply.

The last two lines of the photograph state the autoresonance threshold as

$$\mu > 0.41 \implies \kappa' > 0.41 \sqrt{2} a^{3/4} \simeq 0.58 a^{3/4}. \quad (37)$$

Using the additional handwritten identification $a = \zeta E_0 / (cB_0)$, this is written as

$$\boxed{\kappa' \gtrsim 0.58 \left(\frac{\zeta E_0}{cB_0} \right)^{3/4}.} \quad (38)$$

Transcription note. The numerator in $\xi_E / (cB_0)$ is read provisionally as ξ_E ; its definition is not given on this page. The numerical threshold 0.41 is reproduced from the handwriting. The photograph does not specify its initial-condition assumptions or provide a derivation of that numerical value.